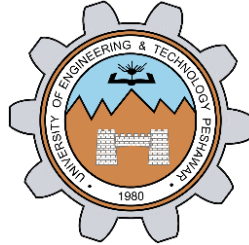


Lab # 05

linked_list_using_Arrays



Spring 2026

Data Structures and Algorithms Lab

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Class Section: **C**

Submitted to:

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Task 01: Implementation of linked_list_using_Arrays:

```
#include <iostream>
using namespace std;

class List
{
private:
    int *arr;
    int capacity;
    int size;

public:
    // Create (Constructor)
    List(int cap)
    {
        capacity = cap;
        size = 0;
        arr = new int[capacity];
    }

    // Add item at given index
    void Add(int index, int value)
    {
        if (size == capacity)
        {
            cout << "List is full!\n";
            return;
        }

        if (index < 0 || index > size)
        {
            cout << "Invalid index!\n";
            return;
        }

        for (int i = size; i > index; i--)
        {
            arr[i] = arr[i - 1];
        }

        arr[index] = value;
        size++;
    }

    // Remove item from given index
    void Remove(int index)
    {
        if (index < 0 || index >= size)
```

```
    {
        cout << "Invalid index!\n";
        return;
    }

    for (int i = index; i < size - 1;
i++)
    {
        arr[i] = arr[i + 1];
    }

    size--;
    }

    // waSize (Returns size of list)
    int waSize()
    {
        return size;
    }

    // IsEmpty
    bool IsEmpty()
    {
        return size == 0;
    }

    // Get item at index
    int Get(int index)
    {
        if (index < 0 || index >= size)
        {
            cout << "Invalid index!\n";
            return -1;
        }

        return arr[index];
    }

    // Start (First element)
    int Start()
    {
        if (IsEmpty())
        {
            cout << "List is empty!\n";
            return -1;
        }

        return arr[0];
    }
}
```

```

}

// End (Last element)
int End()
{
    if (IsEmpty())
    {
        cout << "List is empty!\n";
        return -1;
    }

    return arr[size - 1];
}

// Display List
void Display()
{
    if (IsEmpty())
    {
        cout << "List is empty.\n";
        return;
    }

    cout << "List: ";
    for (int i = 0; i < size; i++)
    {
        cout << arr[i] << " ";
    }
    cout << endl;
}

// Destructor
~List()
{

```

```

        delete[] arr;
    }
};

int main()
{
    List myList(10);

    myList.Add(0, 10);
    myList.Add(1, 20);
    myList.Add(2, 30);
    myList.Add(1, 15);

    myList.Display();

    cout << "Size: " << myList.waSize() <<
endl;
    cout << "Is Empty: " <<
(myList.IsEmpty() ? "Yes" : "No") << endl;
    cout << "Item at index 2: " <<
myList.Get(2) << endl;
    cout << "Start Element: " <<
myList.Start() << endl;
    cout << "End Element: " << myList.End()
<< endl;

    myList.Remove(1);

    cout << "\nAfter Removing index 1:\n";
    myList.Display();

    return 0;
}

```

RUBRICS (LABS/ PROJECTS):

Criteria & Point Assigned	Outstanding 4	Acceptable 3	Considerable 2	Below Expectations 1	Score
Valuing related to work, including responsibility, and professionalism.	Demonstrates excellent responsibility and professionalism in task management. Shows strong commitment to quality work, meeting the deadline, and comprehensive, well-organized effective documentation following all guidelines with exemplary attention to detail.	Demonstrates some responsibility and professionalism. Displays moderate commitment to quality work, meets the deadline and shows sufficient thoroughness in documentation, adhering to guidelines, with some room for enhancement.	Demonstrates limited effort in responsibility and professionalism. Misses the deadline with notable inconsistency in quality work and shows insufficient thoroughness or clarity in documentation, necessitating improvements.	Demonstrates a lack of responsibility and professionalism. Misses the deadline and lacks quality work. Documentation is incomplete, disorganized, or unclear.	
Ability to apply techniques, methods, and procedures effectively.	Executes lab tasks with exceptional skill to design/develop solution and shows proficiency in lab techniques and procedures.	Executes lab tasks with adequate skill to design/develop solution and shows proficiency in lab techniques and procedures.	Executes lab tasks with minimal skill to design/develop solution and shows moderate proficiency in lab techniques and procedures.	Executes lab tasks with limited skill to design/develop solution and lacks proficiency in lab techniques and procedures.	

Demonstrate understanding and relevant Knowledge.	Demonstrates exceptional understanding of concepts with clear explanation of investigation and analysis of the objectives and outcomes.	Demonstrates a good understanding of concepts with some explanations of investigation and analysis that align with the lab's objectives and outcomes.	Demonstrates a basic understanding of key concepts but lacks clarity or depth in explaining the investigation and analysis, with some inconsistencies.	Demonstrates limited understanding of concepts with unclear explanations of investigation and analysis that do not align with the lab's objectives and outcomes.
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